

Soner Albayrak

Middle East Technical University
Ankara 06800
Turkey

Email : contact@soneralbayrak.com
Phone : +90-312-210-3281
Web: soneralbayrak.com

ACADEMIC POSITIONS

Middle East Technical University, Ankara, Turkey	
Doktor Öğretim Üyesi (Assistant Professor)	Oct 2023 - present
Ek Görevli Öğretim Görevlisi (Part Time Lecturer)	Sep 2023 - Oct 2023
University of Amsterdam, Amsterdam, Netherlands	Sep 2021 - Sep 2023
Onderzoeker (Researcher)	(Contact: Daniel Baumann)
National Taiwan University, Taipei, Taiwan	Apr 2023 - Jun 2023
Visiting Researcher	(Contact: Yu-tin Huang)

ACADEMIC AND PROFESSIONAL PREPARATION

Yale University, New Haven, USA	Aug 2016 - Jun 2021
Doctor of Philosophy in Physics	(Advisor: David Poland)
Dissertation title: <i>Analytic Studies of Fermions in the Conformal Bootstrap</i>	
Master of Philosophy in Physics	
Master of Science in Physics	
California Institute of Technology, Pasadena, USA	Jul 2018 - Jun 2019
Visiting Student Researcher	(Host: David Simmons Duffin)
Simons Bootstrap Collaboration Graduate Student Exchange	
Middle East Technical University, Ankara, Turkey	Sep 2008 - Feb 2016
Master of Science in Physics	(Advisor: İsmail Turan)
Thesis: <i>A Lorentz Violating Theory: Its Nonminimal Extension in the Photon Sector</i>	
Bachelor of Science in Physics	
Bachelor of Science in Electrical & Electronics Engineering	

CURRENT RESEARCH INTERESTS

Conformal bootstrap	AdS/CFT
Cosmological bootstrap	Scattering amplitudes

PEER REVIEWED PUBLICATIONS

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- S. Albayrak** and S. Kharel, *Soft limits of gluon amplitudes in holography and cosmology*, [JHEP 07 \(2025\) 019](#)
- S. Albayrak**, S. Kharel and X. Wang, *Momentum-space formulae for AdS correlators for diverse theories in diverse dimensions*, [JHEP 07 \(2024\) 281](#)
- S. Albayrak**, P. Benincasa and C. D. Pueyo, *Perturbative unitarity and the wavefunction of the Universe*, [SciPost Phys. 16, 157 \(2024\)](#)
- S. Albayrak** and S. Kharel, *All plus four point (A)dS graviton function using generalized on-shell recursion relation*, [JHEP 05 \(2023\) 151](#)
- S. Albayrak**, Rajeev Erramilli, Zhijin Li, David Poland and Yuan Xin, *Bootstrapping $N_f = 4$ conformal QED₃*, [Phys.Rev.D 105 \(2022\)](#)
- S. Albayrak**, S. Kharel and D. Meltzer, *On duality of color and kinematics in (A)dS momentum space*, [JHEP 03 \(2021\) 249](#)

- S. Albayrak** and S. Kharel, *Spinning loop amplitudes in Anti-de Sitter space*, [Phys.Rev.D 103 \(2021\) 2](#)
- S. Albayrak**, C. Chowdhury and S. Kharel, *On loop celestial amplitudes for gauge theory and gravity*, [Phys.Rev.D 102 \(2020\)](#)
- S. Albayrak**, D. Meltzer and D. Poland, *The Inversion Formula and 6j Symbol for 3d Fermions*, [JHEP 09 \(2020\) 148](#)
- S. Albayrak**, C. Chowdhury and S. Kharel, *Study of momentum space scalar amplitudes in AdS spacetime*, [Phys.Rev.D 101 \(2020\) 12](#)
- S. Albayrak** and S. Kharel, *Towards the higher point holographic momentum space amplitudes. Part II. Gravitons*, [JHEP 12 \(2019\) 135](#)
- S. Albayrak**, C. Chowdhury and S. Kharel, *New relation for Witten diagrams*, [JHEP 10 \(2019\) 274](#)
- S. Albayrak**, D. Meltzer and D. Poland, *More Analytic Bootstrap: Nonperturbative Effects and Fermions*, [JHEP 08 \(2019\) 040](#)
- S. Albayrak** and S. Kharel, *Towards the higher point holographic momentum space amplitudes*, [JHEP 02 \(2019\) 040](#)

TEACHING EXPERIENCE

Instructor , Middle East Technical University	2023-current
PHYS 778: <i>Bootstrap Methods in Theoretical Physics II</i>	Spring '24
PHYS 777: <i>Bootstrap Methods in Theoretical Physics I</i>	Fall '24
PHYS 332: <i>Electromagnetic Theory II</i>	Spring '25
PHYS 331: <i>Electromagnetic Theory I</i>	Fall '25
PHYS 210: <i>Mathematical Methods in Physics II</i>	Spring '23, '24
PHYS 209: <i>Mathematical Methods in Physics I</i>	Fall '23, '24
PHYS 106: <i>General Physics II</i>	Spring '23, '25
PHYS 105: <i>General Physics I</i>	Fall '23, '25
Teaching Assistant , Yale University	2017-2021
PHYS 181: <i>University Physics</i>	Spring '18, '20, '21
PHYS 180: <i>University Physics</i>	Fall '17, '19, '20; Summer '20
PHYS 166L: <i>General Physics Laboratory</i>	Spring '17
Teaching Assistant , Middle East Technical University	2014-2016
PHYS 502: <i>Analytical Mechanics</i>	Fall '15; Spring '16
PHYS 408: <i>Particle Physics II</i>	Spring '15, '16
PHYS 407: <i>Particle Physics I</i>	Fall '14, '15
PHYS 311: <i>Classical Mechanics</i>	Fall '14, '15; Spring '15, '16
PHYS 105 - LAB: <i>General Physics I</i>	Fall '14

AWARDS, HONORS, SCHOLARSHIPS, AND FELLOWSHIPS

The Science Academy Young Scientist Award (BAGEP 2024) <i>Durmuş Ali Demir Anısına BAGEP Anı Ödülü</i>	Apr 2024
International Fellowship for Early Stage Researchers Program (TÜBİTAK 2232b)	Oct 2023
Graduate Courses Performance Award <i>by Middle East Technical University</i>	Apr 2015

National Graduate Study Scholarship by Scientific and Technological Research Council of Turkey	Oct 2014
Dean's Honor and High Honor Lists	2009-2013
Outstanding Success Scholarship's by METU, by Turkish Educational Foundation, and by Ministry of Youth and Sports	Aug 2008

CONFERENCES, PRESENTATIONS, AND PROCEEDINGS

42. International Physics Congress, Herodot Cultural Center - Bodrum Talk: <i>Harmonic Analysis of Bosonic Conformal Field Theories</i>	Sep 2026
Bootstrap 2026, London	Aug 2026
6. Yüksek Enerji Fiziği, Astrofizik ve Kozmoloji Çalıştayı (YEFAK 2026), Yıldız Teknik Üniversitesi - İstanbul Invited Talk: <i>Soft limits in holography and cosmology</i>	June 2026
ODTÜ 70. Yıl Dönem Arası Seminerleri, METU - Ankara Invited Talk: <i>Şu Fiziği İndirgeyip de mi Kuramlasak Yoksa İndirgemeyip de mi Kuramlasak?</i>	Feb 2026
AGEP-11 Kapanış Programı, METU NCC - Cyprus Invited Talk: <i>Teorik Fizikteki Problemler</i>	Nov 2025
41. International Physics Congress, Herodot Cultural Center - Bodrum Talk: <i>Soft Limits of Gluon Amplitudes in Holography and Cosmology</i> Talk & Full-text proceeding: <i>A Geometric Approach to the Perturbative Unitarity in Cosmology</i>	Sep 2025
Bootstrap 2025, ICTP – SAIFR	Jul 2025
TEDx, ODTÜ Geliştirme Vakfı Lisesi - Ankara Invited Talk: <i>Navigating the Frontiers of Knowledge</i>	May 2025
Departmental Seminar, Bilkent University - Ankara Invited Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	Apr 2025
1.Fizik ve Felsefe Çalıştayı, İstanbul University Talk: <i>Şu Fiziği indirgeyip de mi kuramlasak yoksa indirgemeyip de mi kuramlasak?</i>	Apr 2025
4. ODTÜ Fizik Topluluğu Çalıştayı, METU - Ankara Invited Talk: <i>Teorik Fizikteki Problemler</i>	Mar 2025
Departmental Seminar, Bosphorus University - İstanbul Invited Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	Feb 2025
A life dedicated to theoretical physics: in memory of Durmuş Ali Demir, Sabancı University - İstanbul Invited Talk: <i>A geometric approach to the perturbative unitarity in cosmology</i>	Feb 2025
StarFin 2024 Bilim Kongresi, online seminar Invited Talk: <i>Teorik Fizikteki Problemler</i>	Dec 2024
Bootstrap 2024, Complutense University	Jul 2024
Yüksek Enerji Fiziği Çalıştayı (YEFİST 2024), İstanbul University - İstanbul Talk: <i>Problems in Theoretical Physics</i>	May 2024
21 th Workshop on Quantization, Dualities and Integrable Systems, online seminar Talk: <i>Perturbative Unitarity and the Wavefunction of the Universe</i>	Apr 2024
Astronomi ve Fizik Günleri, Bosphorus University - İstanbul Invited Talk: <i>Problems in Theoretical Physics</i>	Apr 2024
Conformal Bootstrap Annual Meeting, Simons Foundation - New York	Nov 2023

Fuzzy Sphere Meets Bootstrap, SCGP-Stony Brook	Nov 2023
Bootstrap 2023, ICTP – SAIFR	Jul 2023
National Taiwan University, seminar	May 2023
Invited Talk: <i>A geometric approach to the perturbative unitarity in cosmology</i>	
Middle East Technical University, seminar	Mar 2023
Invited Talk: <i>Extending the flat space techniques to curved spacetimes</i>	
University of Amsterdam String Theory Journal Club	Feb 2023
Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	
Positivity, Princeton Center for Theoretical Science	Dec 2022
Bootstrapping Nature, Galileo Galilei Institute	Nov 2022
Invited Talk: <i>Inverting the 3d Fermions</i>	
ATITPhysics 2022S, online school	Sep 2022
Colloquium: <i>Problems in Theoretical Physics</i>	
Middle East Technical University, online seminar	Jun 2022
Invited Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	
20 th Workshop on Quantization, Dualities and Integrable Systems, online seminar	Apr 2022
Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	
Holography@D-ITP meetings, Utrecht University	Apr 2022
Invited Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	
Case Western Reserve University, online seminar	Apr 2022
Invited Talk: <i>Conformal QED in 3d: The Numerical Bootstrap Approach</i>	
Boundaries and Defects in CFT and Holography, online school	Mar 2022
Conformal Bootstrap Annual Meeting, Simons Foundation - New York	Nov 2021
ICTP – School on Superstring Theory and Related Topics, online school	Oct 2021
University of Amsterdam String Theory Journal Club	Oct 2021
Talk: <i>Fermions in CFTs: The Analytic Bootstrap Approach</i>	
International Society for Quantum Gravity, online meeting	Oct 2021
All Lambdas Holography @ Prague, online workshop	Oct 2021
ATITPhysics 2021, online school	Sep 2021
Lecture Series: <i>Conformal Symmetry in Field Theories</i>	
Bootstrap 2021, online conference	Jul 2021
19 th Workshop on Quantization, Dualities and Integrable Systems, online seminar	Apr 2021
Talk: <i>Fermions in CFTs: The Analytic Bootstrap Approach</i>	
THK University, online seminar	Mar 2021
Invited Talk: <i>Simetriler: Niçin evrenimiz böyle?</i>	
Porto University, online seminar	Feb 2021
Invited Talk: <i>On Witten diagrams in momentum space</i>	
Bootstrap 2020, online conference	Jun 2020
Middle East Technical University, online seminar	Nov 2020
Invited Talk: <i>Recent Developments in Conformal Bootstrap</i>	
Zoomplitudes 2020, online conference	May 2020
Online seminar series in theoretical physics in Turkey	May 2020
Talk: <i>Recent Developments in Conformal Bootstrap</i>	

Conformal Bootstrap Annual Meeting, Simons Foundation - New York	Nov 2019
Developments in the Numerical Bootstrap, SCGP-Stony Brook	Nov 2019
Bootstrap 2019, Perimeter Institute	Jul 2019
Bootstrap 2018, California Institute of Technology	Jul 2018
Analytic approaches to the bootstrap, Azores	May 2018
Particle theory group student talk: <i>On Wess-Zumino-Witten model</i> , Yale	Mar 2018
Signals for Nonminimal Lorentz and CPT Violation, IUCSS	Jun 2015
Poster presentation: <i>CPT-odd Photon in Vacuum-Orthogonal Model</i>	
Ankara High Energy Physics Days, METU	Feb 2015
Talk: <i>CPT Violating QED and Beyond</i>	
Recent Developments in High Energy Physics, METU	May 2013

OTHER ACTIVITIES

Active

Referee, <i>İstanbul Commerce University Journal of Science</i>	2025-current
Referee, <i>Turkish Journal of Physics</i>	2024-current
Referee, <i>Journal of High Energy Physics</i>	2023-current
Member,	2023-current

Past

Organizer, <i>BOĞAZİÇİ-METU-OSTIMTECH Feza Gürsey Commemoration Day</i>	2025
Member, <i>University Forum (UVA)</i> <i>served on the representative group for postdoctoral researchers</i>	2022-2023
Organizer, <i>ATITPhysics Online School Series</i>	2021-2022
Advising Committee, <i>Joint Seminar Series (Boun&METU&UTAA)</i>	2021-2022
Physics Outreach (in Turkish), <i>Evrin Ağacı</i>	2017-2018
Electrical & Electronics Engr. Intern, PROMOD R&D Software, Ankara <i>worked for object tracking and camera calibration problems</i>	Jul & Aug 2013
Electrical & Electronics Engr. Intern, SÖKTAŞ Textile, Aydın <i>worked in the control department</i>	Aug & Sep 2012

REFERENCES

Available upon request